A Distribution of Possibilities

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for A Distribution of Possibilities before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported A Distribution of Possibilities trace and record the first step you can explain without help.

Use W2: Calculate the paths for Next token **distribution**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed A Distribution of Possibilities explanation against the supplied facts before discussing it.

Use the worked case for Next token **distribution**s. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Analyse a new table independently

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the A Distribution of Possibilities target to a trace and a justified conclusion.

> A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

Solve the given case

> Solve the supplied A Distribution of Possibilities case and show the deciding evidence.

A toy next-token **distribution** is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A toy next-token **distribution** assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 2 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	

Test record	My evidence
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what A Distribution of Possibilities adds to Verified Study Buddy and keep the evidence.

Explain that Study Buddy responses come from possibilities, so important claims need checks.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Validate the table

> Validate the table. Apply the supplied A Distribution of Possibilities facts and keep your reasoning.

Check 0.5,0.3,0.2 as probabilities. Interpret their sum and the complement of ready.

My reasoning and evidence

W2 Calculate the paths

> Calculate the paths. Apply the supplied A Distribution of Possibilities facts and keep your reasoning.

Compute ready,end; busy,end; unknown,end and their combined probability.

My reasoning and evidence

W3 Test a greedy claim

> Test a greedy claim. Apply the supplied A Distribution of Possibilities facts and keep your reasoning.

Use A=0.6,B=0.4 and end conditional probabilities 0.51 and 1. Compare greedy A,end with B,end.

My reasoning and evidence

W4 Analyse a new table independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Analyse a new table independently. Apply the supplied A Distribution of Possibilities facts and keep your reasoning.

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Normalise with a stated rule

> Normalise with a stated rule. Apply the supplied A Distribution of Possibilities facts and keep your reasoning.

Weights are ready 6,busy 3,unknown 1. Convert them into probabilities. What if a weight is -1?

My reasoning and evidence

WORKBOOK · 17 OF 26

Embeddings and position — W1 Look up vectors

> Use the supplied A Distribution of Possibilities task to record your reasoning, evidence and next step.

For red blue name the IDs, **embedding** rows and shape before adding positions.

My reasoning and evidence

WORKBOOK · 18 OF 26

Embeddings and position — W2 Add positions

Use the supplied A Distribution of Possibilities task to record your reasoning, evidence and next step.

Compute representations for red blue and blue red under the stated position table.

My reasoning and evidence

WORKBOOK · 19 OF 26

Embeddings and position — W3 Audit the sum

Use the supplied A Distribution of Possibilities task to record your reasoning, evidence and next step.

Sum the rows for both sequences. Does the sum establish that the original text had the same order?

My reasoning and evidence

WORKBOOK · 20 OF 26

Embeddings and position — W4 Trace new vectors independently

Use the supplied A Distribution of Possibilities task to record your reasoning, evidence and next step.

Embeddings A=[1,2], B=[3,0], position 0=[0,1], position 1=[1,0]. Compute represented rows and sums for A B and B A. Explain what the equal sums mean.

My reasoning and evidence

WORKBOOK · 21 OF 26

Embeddings and position — W5 Handle a shape failure

> Use the supplied A Distribution of Possibilities task to record your reasoning, evidence and next step.

An **embedding** has width two but its **position vector** has width three. The **input** also contains a third position absent from the table. Specify honest handling.

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: A Distribution of Possibilities

> A fictional two-stage token tree illustrates next-token prediction. Explain the labelled step before checking the changed condition.

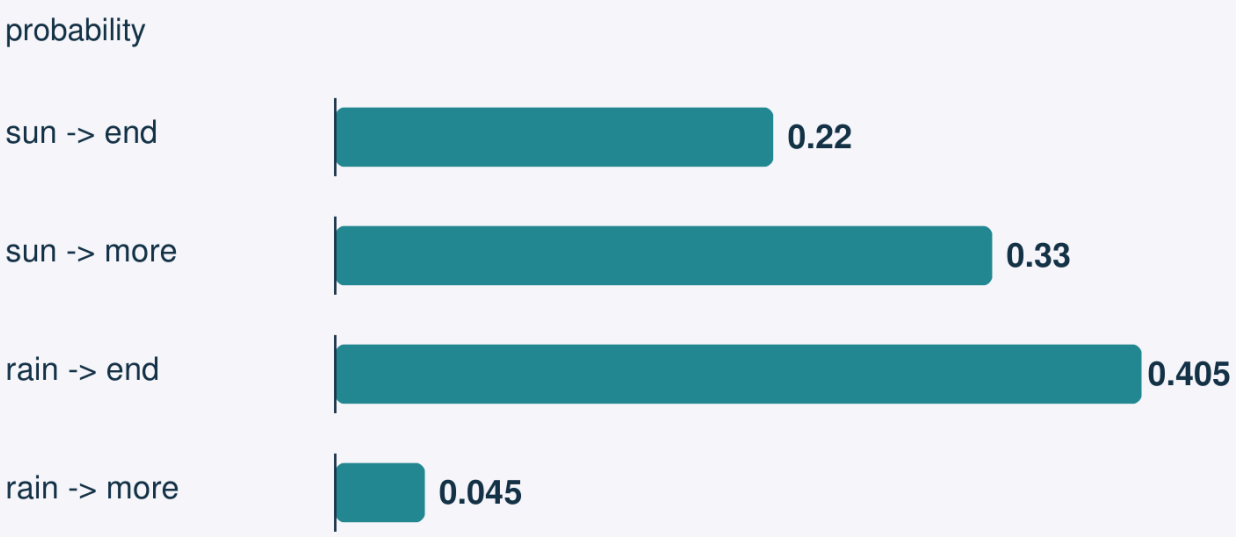
A fictional two-stage token tree illustrates next-token **prediction**.

First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

AI ACADEMY / AI-3/2

Joint probabilities of the four toy branches



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/2

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end:0.55 x 0.4=0.22; sun -> more:0.55 x 0.6=0.33.



3. Rain branches

rain -> end:0.45 x 0.9=0.405; rain -> more:0.45 x 0.1=0.045.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is 0.625 not a broken probability sum?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional two-stage token tree illustrates next-token **prediction**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? rain→end is the more probable of the two complete end paths.

> 1. Rain branches

2. Compare end paths

3. Sun branches

4. First **distribution**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare end paths rain→end is the more probable of the two complete end paths. Conditional probabilities describe what follows a particular prefix.

Multiplying along a path gives its joint probability. The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended.

Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. sun→end becomes 0.44, beating unchanged rain→end 0.405.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: sun→end becomes 0.44, beating unchanged rain→end 0.405.

> Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Repair a missing building block

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

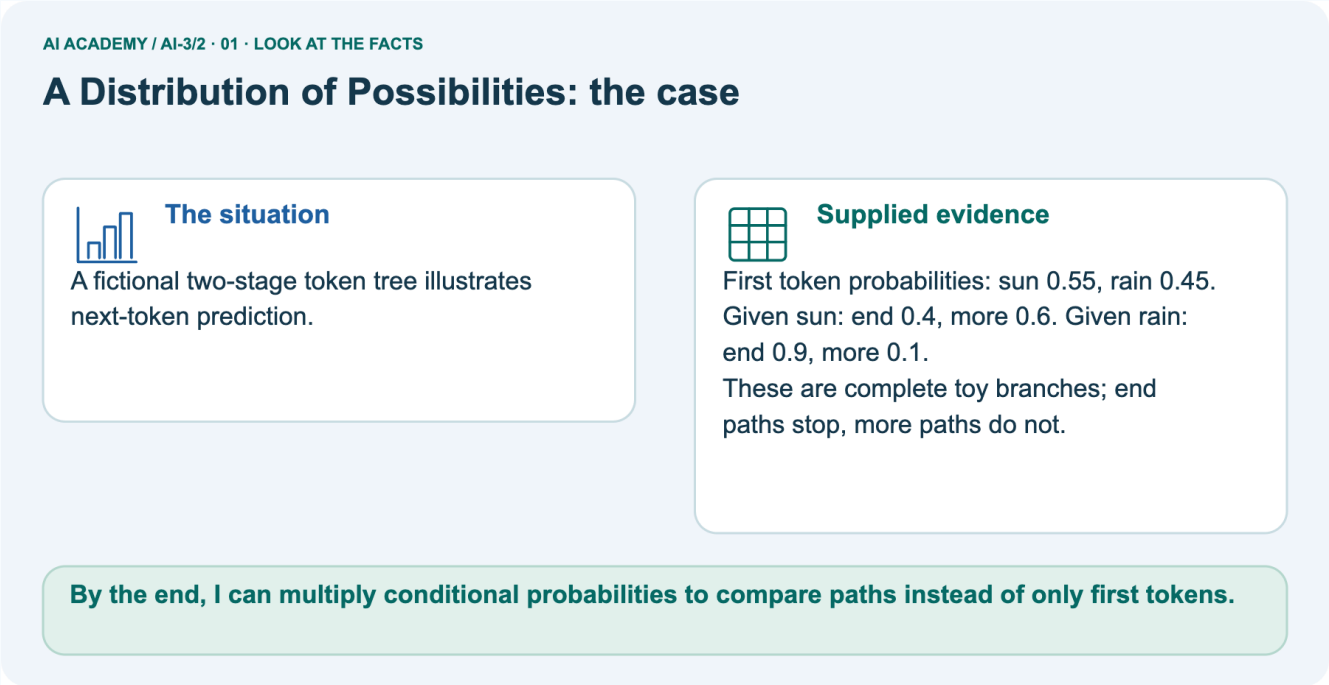
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each A Distribution of Possibilities picture from its labels, then record a relevant limit.

01. A Distribution of Possibilities: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

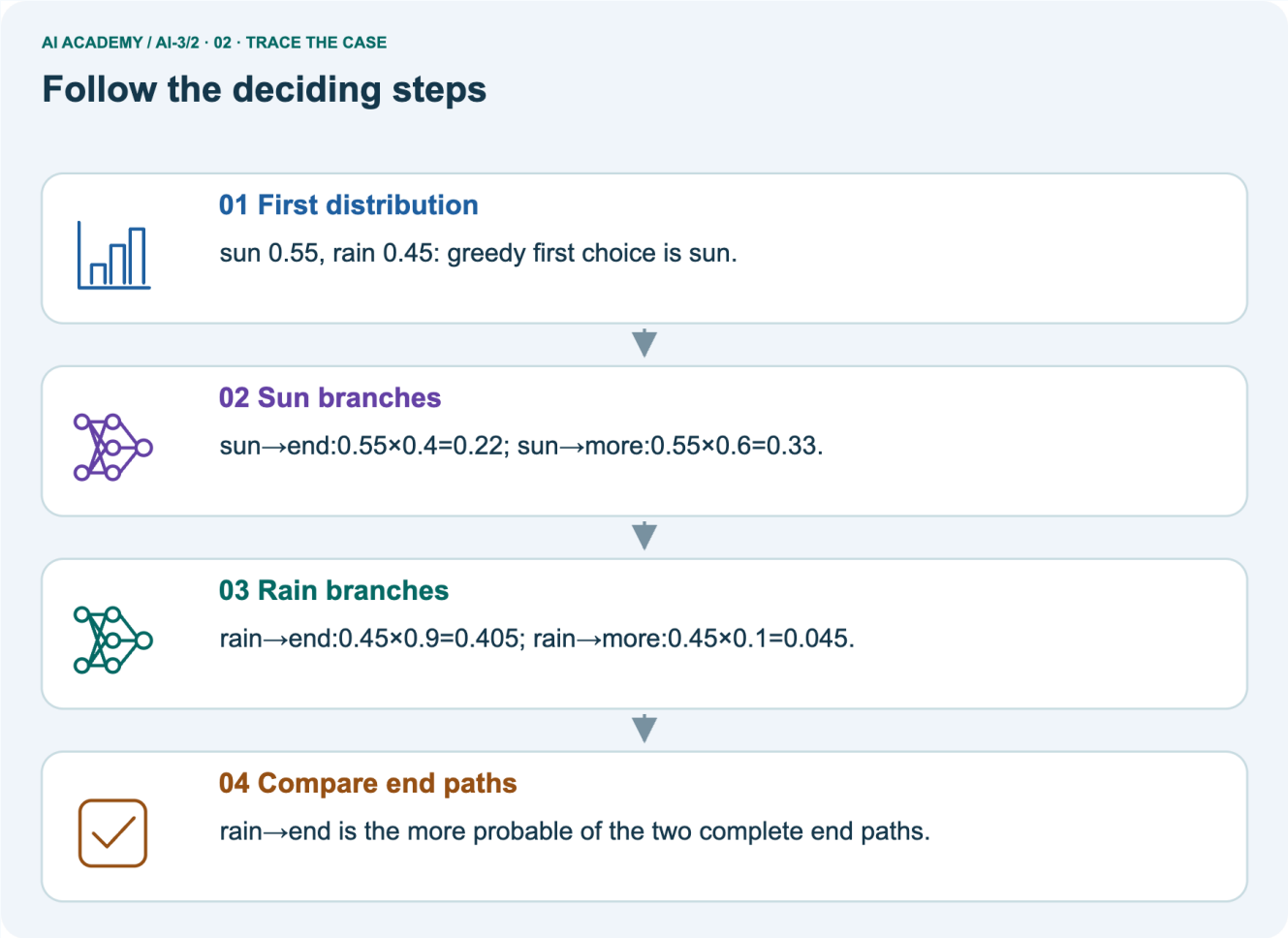
A fictional two-stage token tree illustrates next-token **prediction**. Supplied facts: First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



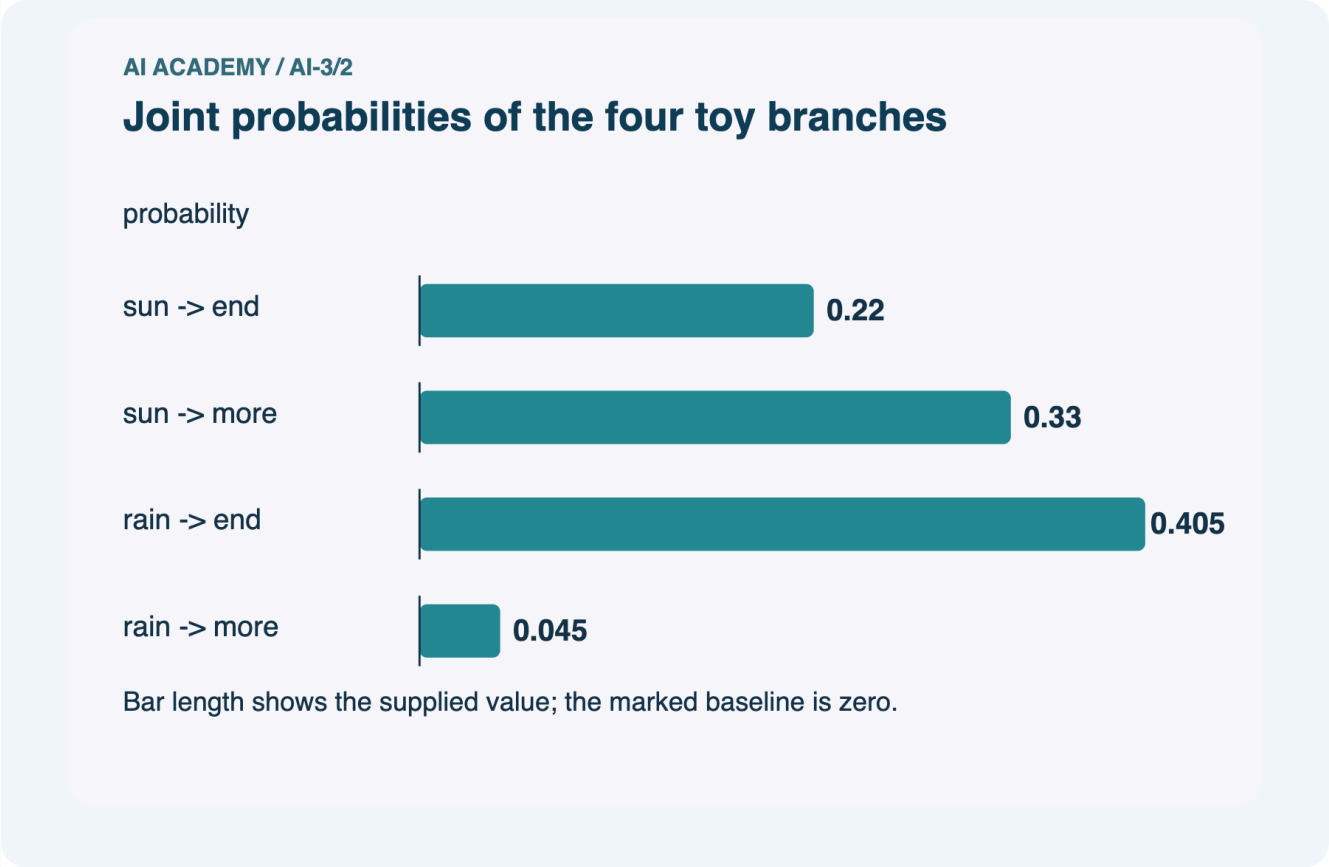
Follow the labelled picture with the matching explanation. Open the vector diagram

First **distribution**: sun 0.55, rain 0.45: greedy first choice is sun. → Sun branches: sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33. → Rain branches: rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045. → Compare end paths: rain→end is the more probable of the two complete end paths.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Joint probabilities of the four toy branches



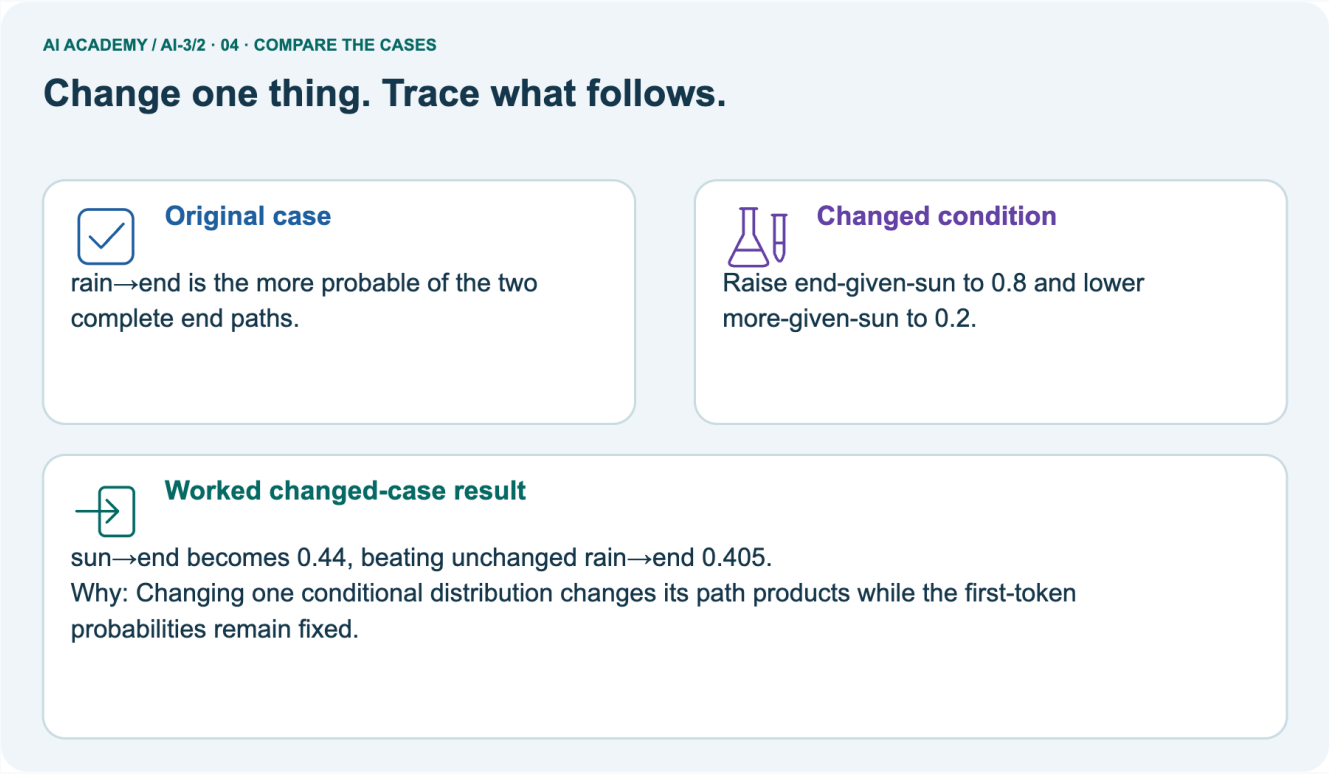
Follow the labelled picture with the matching explanation. Open the vector diagram

Joint probabilities of the four toy branches. Units: probability. sun→end: 0.22. sun→more: 0.33. rain→end: 0.405. rain→more: 0.045.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: rain→end is the more probable of the two complete end paths. Changed condition: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

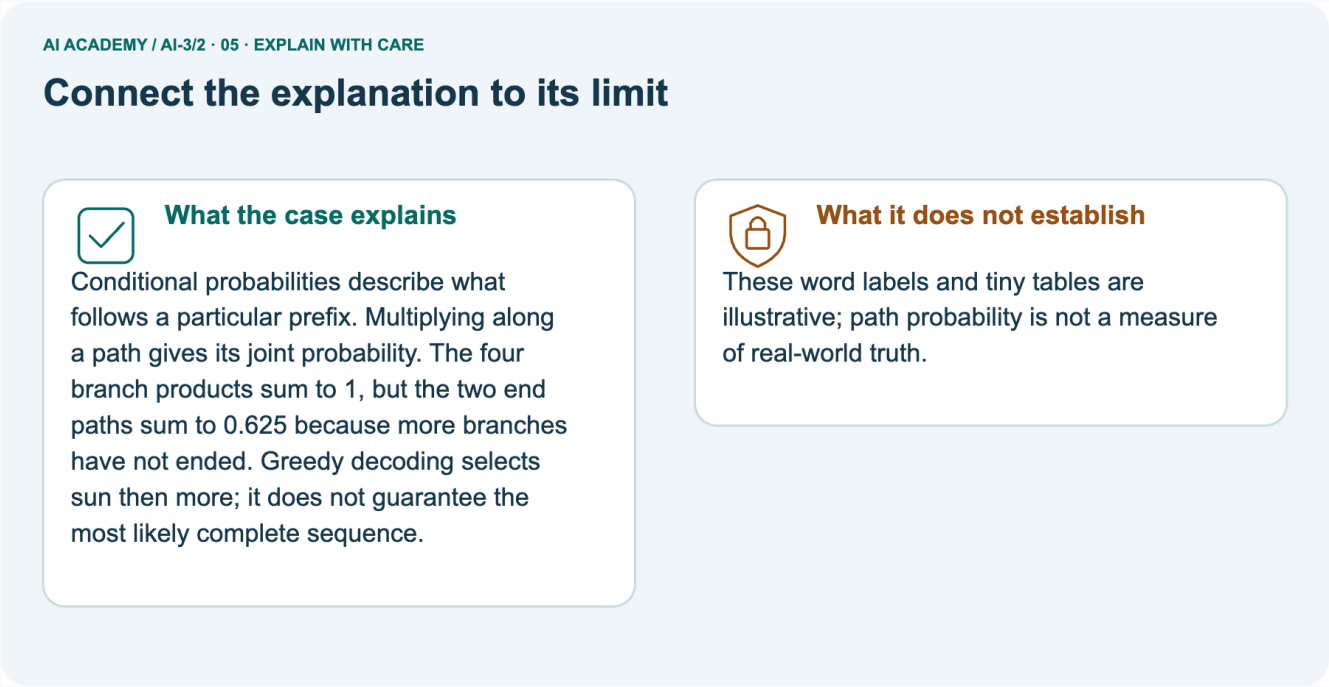
Expected result: sun→end becomes 0.44, beating unchanged rain→end 0.405. Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Predict before checking the worked changed-case result. Name what changed and what the

supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

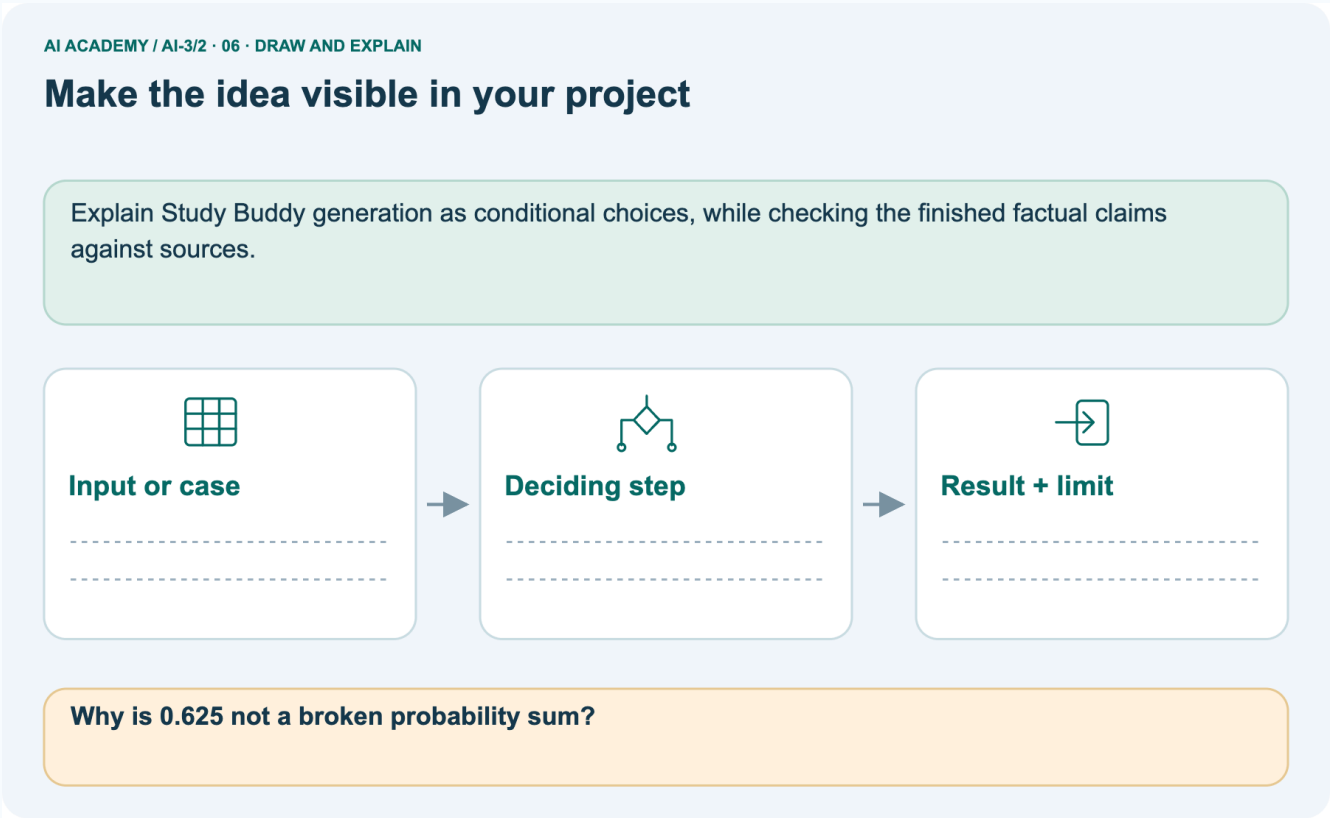
The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Limit: These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources. Draw your **input**, deciding step, result and limit.

Supported question: Why is 0.625 not a broken probability sum?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.